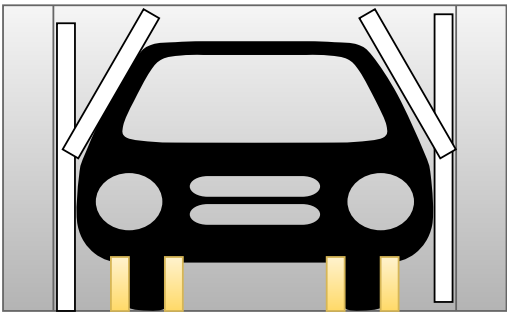
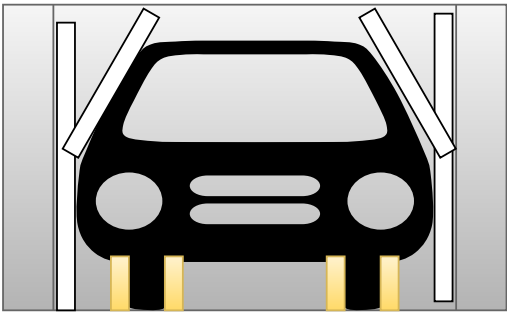
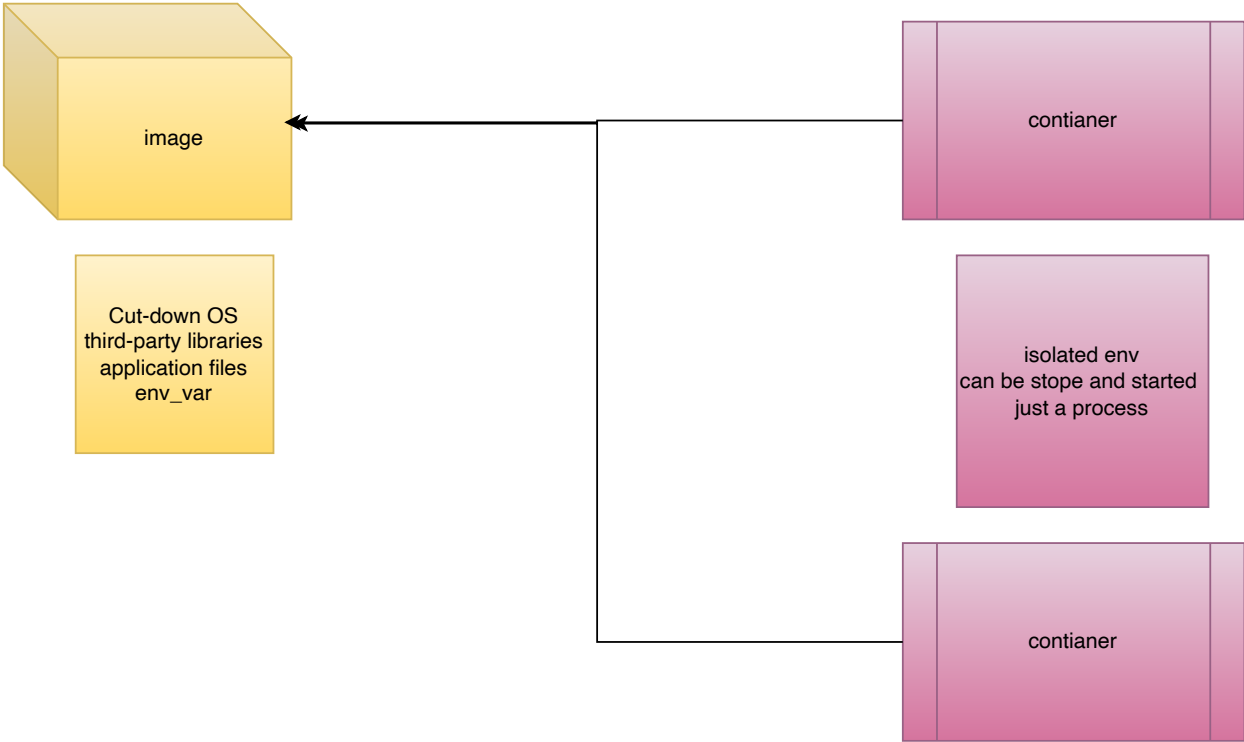


Docker images



shipping agreement

contianer must be in a black color
the wheels must be tied to the ground
the wheels must be at 30 degree form the door
the door must be locked

~~~~~  
~~~~~  
~~
~

docker

why docker ?

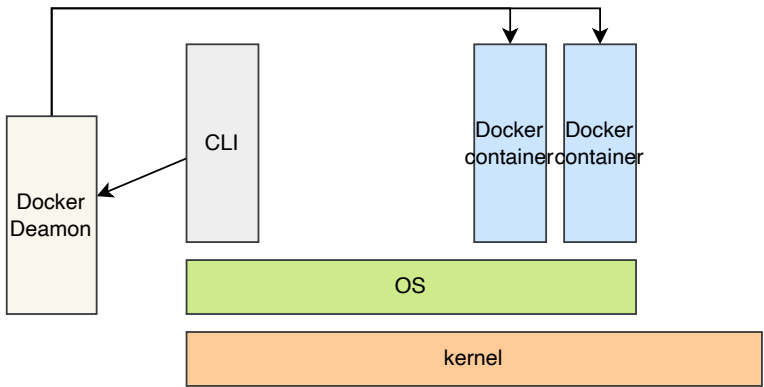
- consistency : works on every machine
- isolation: application run in isolated env
- efficiency : lightweight os
- scalability : ease to scale

reasons:

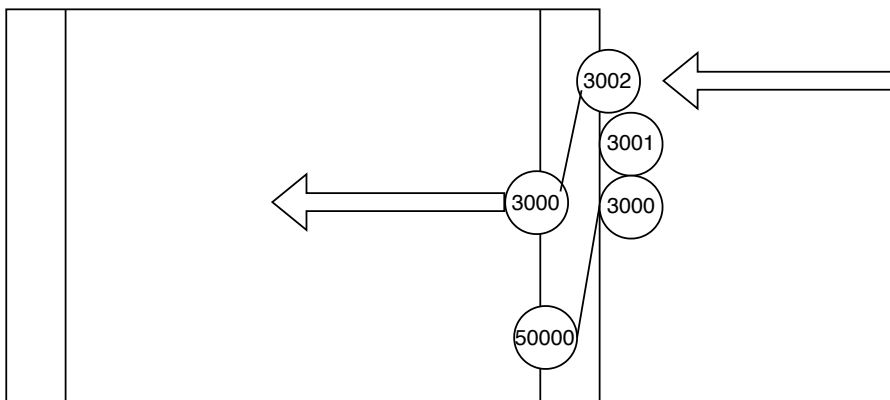
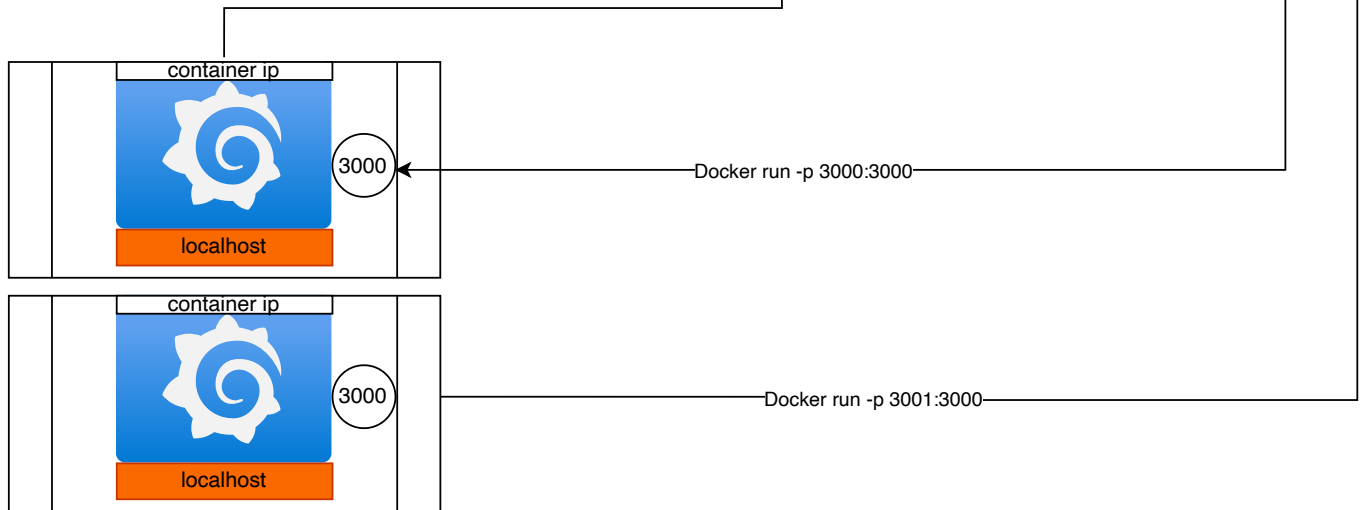
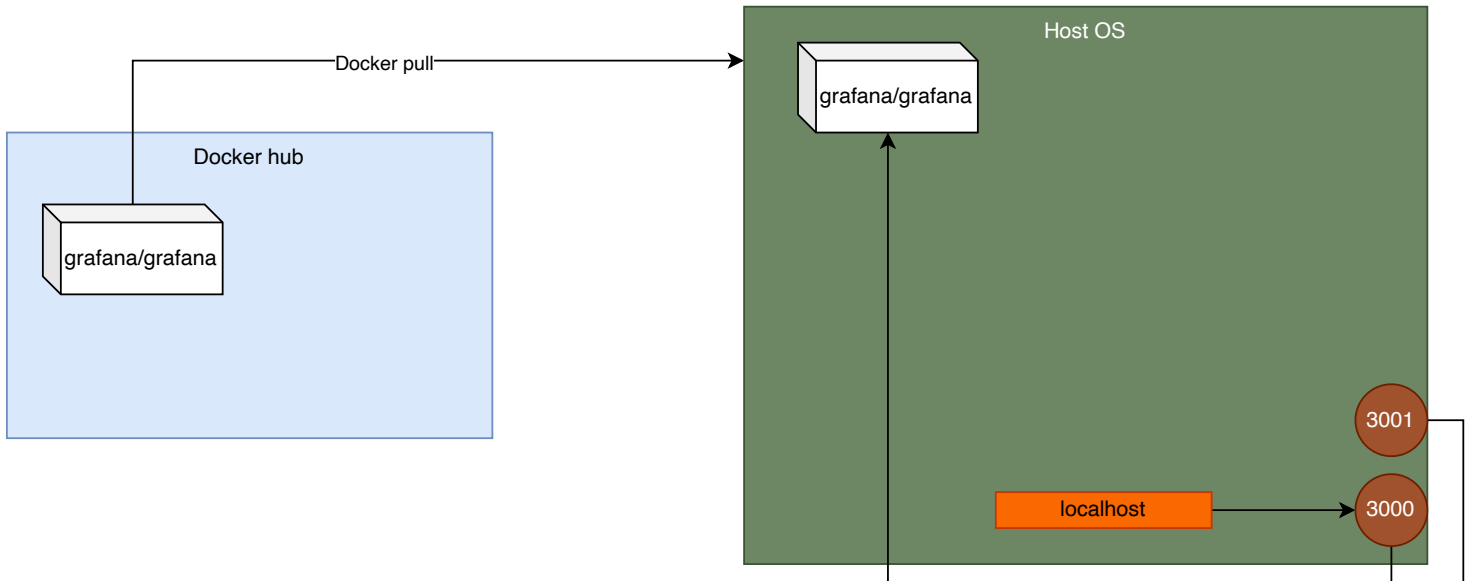
- one or more files missing in the prod
- software version mismatch
- different configuration setting

Docker Archithcture:

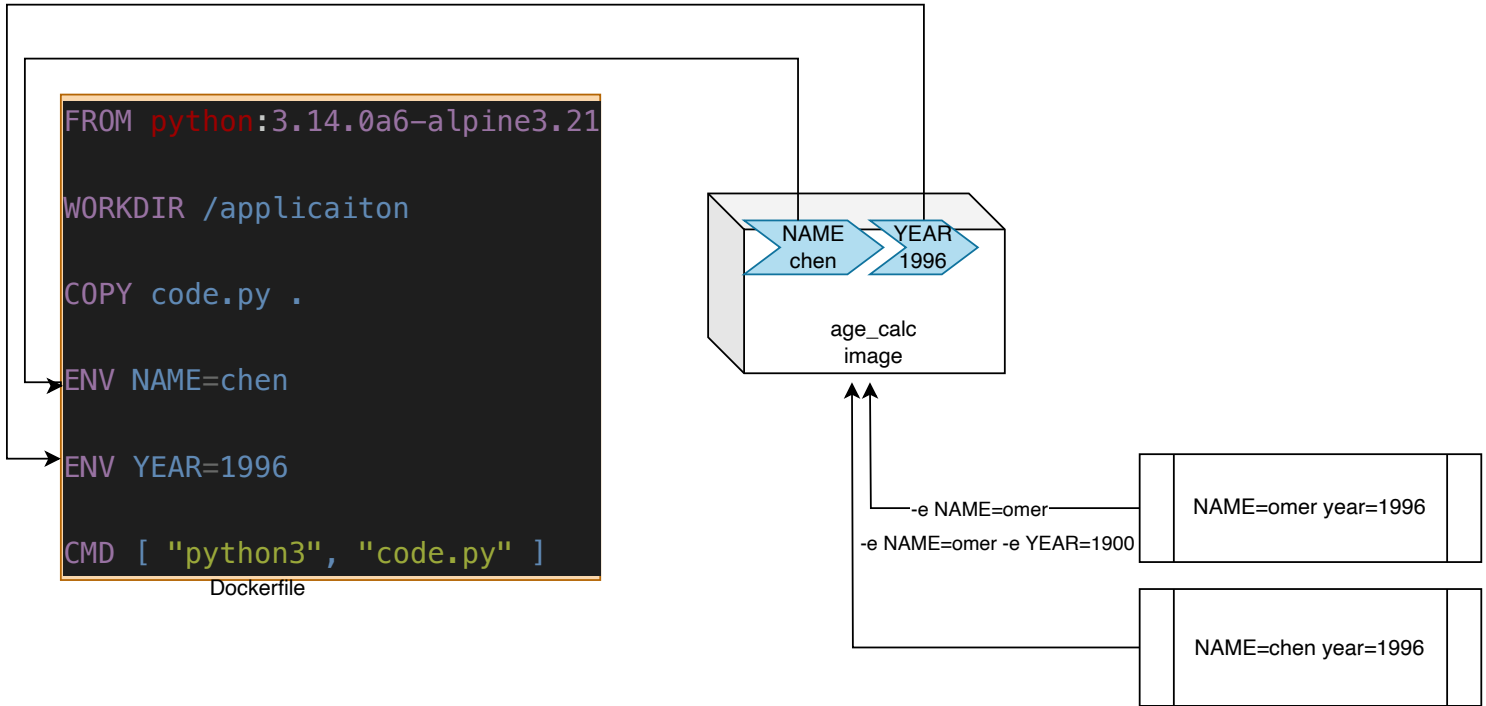
docker object
containers networks
volumes images

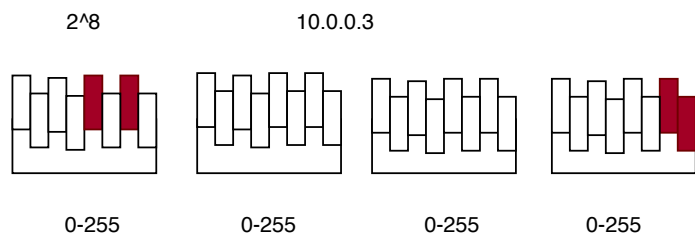
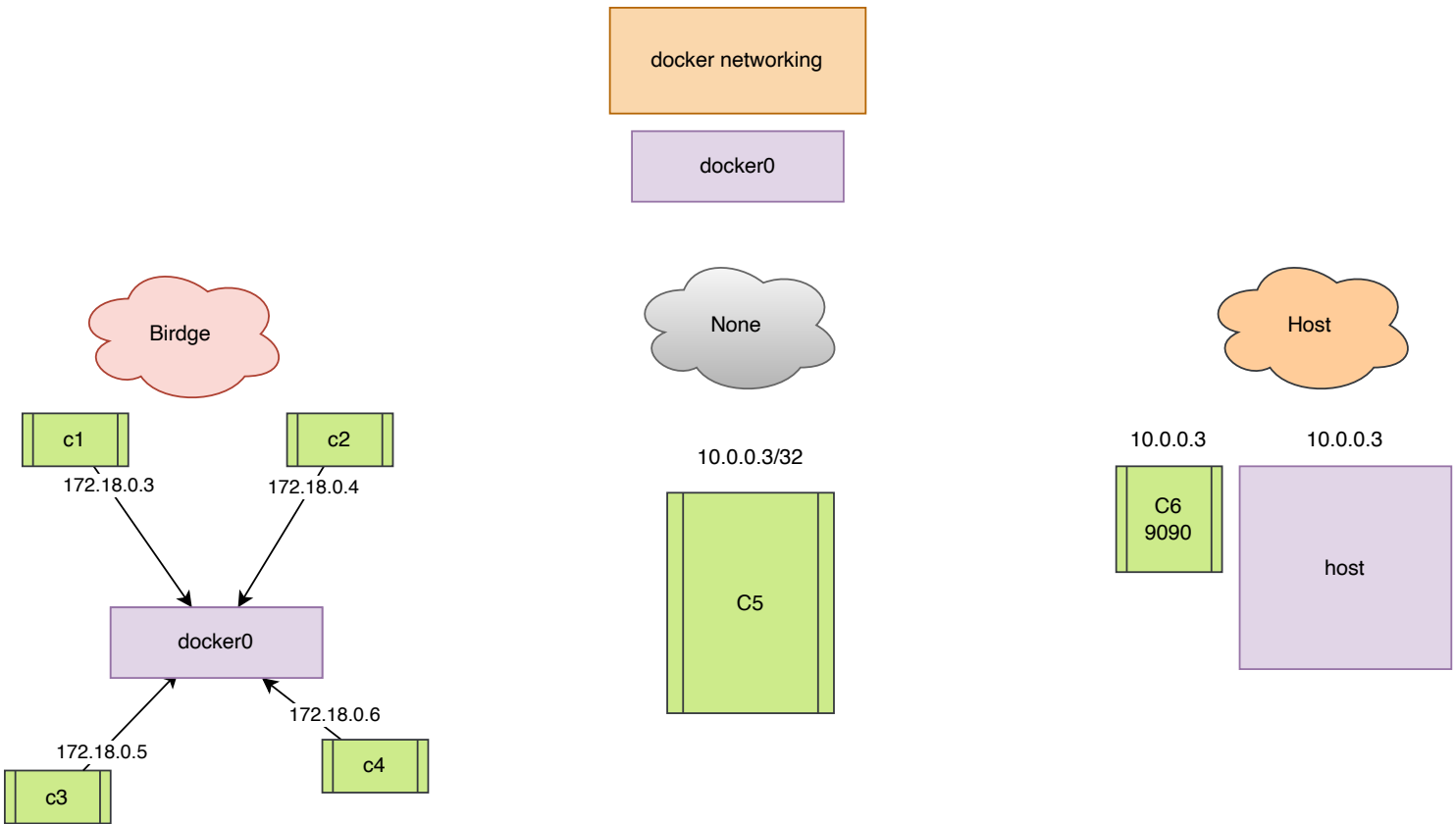


Running containers

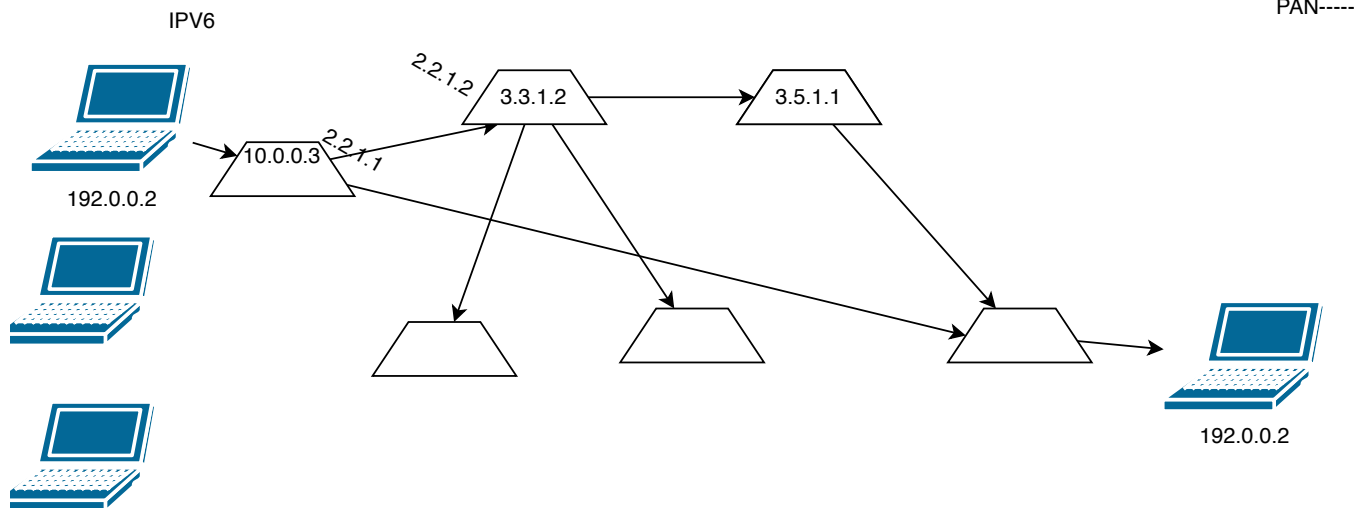


environment variables in
docker





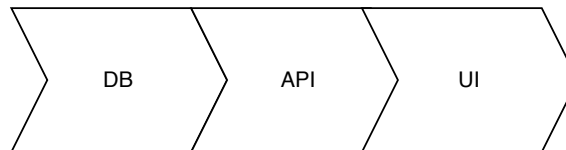
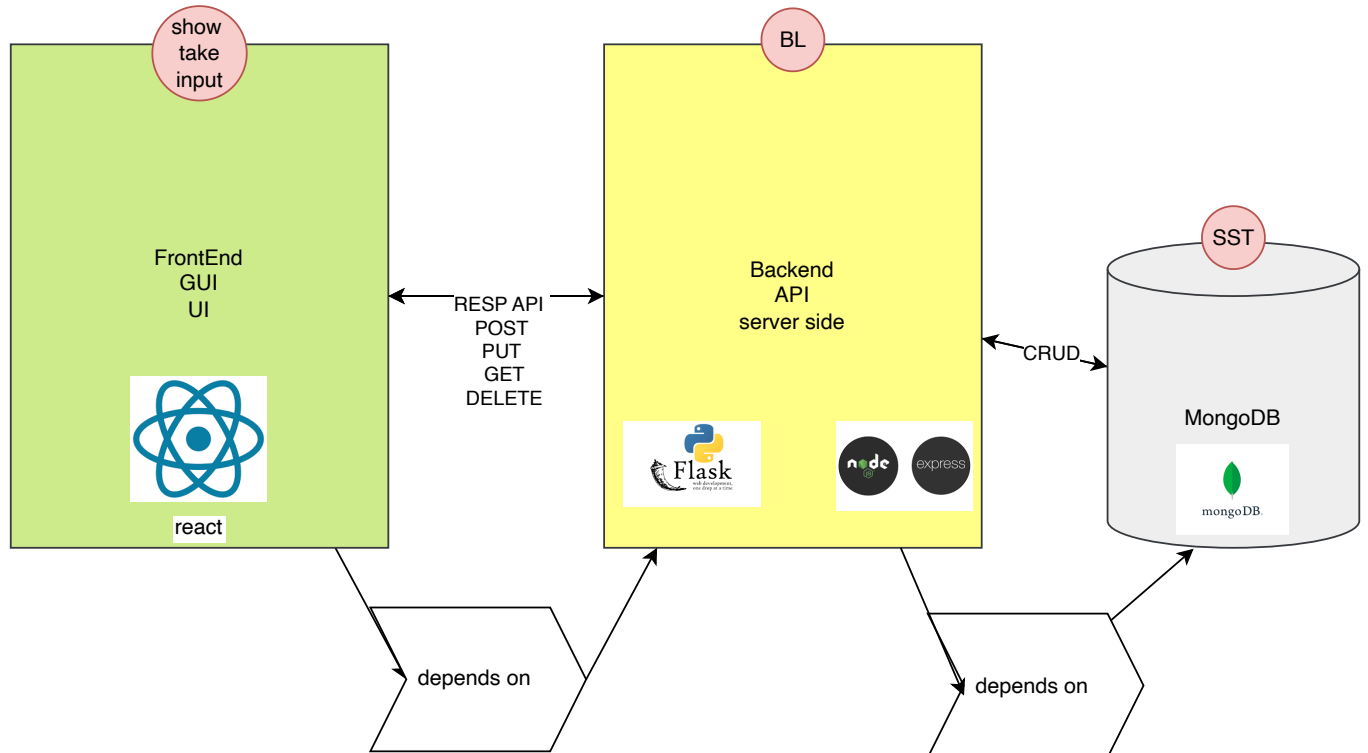
$256 \times 4 = 1,024$



Docker full exercise

https://github.com/hothaifa96/DevSecOps18/tree/docker/class_code/docker_networks

TASK MANAGER APPLICATION



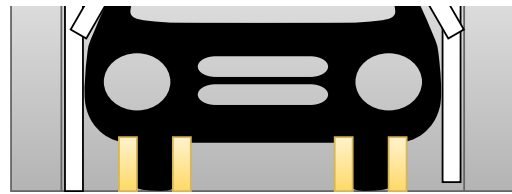
start a new mongodb container

27017

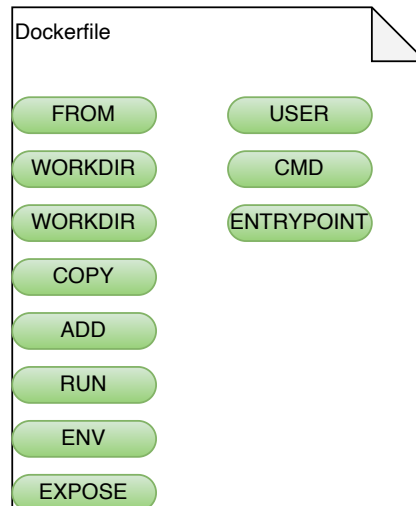
publish port

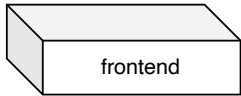
network tast-m

network
tast-m

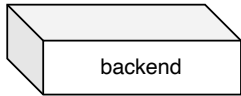


- OS + python env
- folder
- code files
- install 3rd party lib
- keep port open
- how to run the application

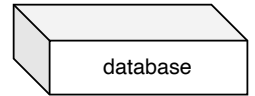




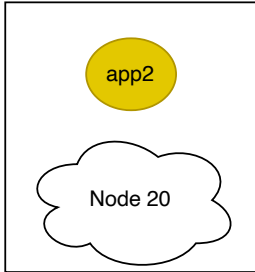
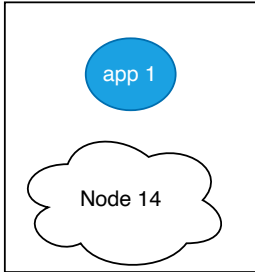
REACT
Node 14



javascript
Node 20



postgresql
13



Virtual amachines



hypervisor



hypervisor



hypervisor

vmware,virtualbox

Contianer

windows

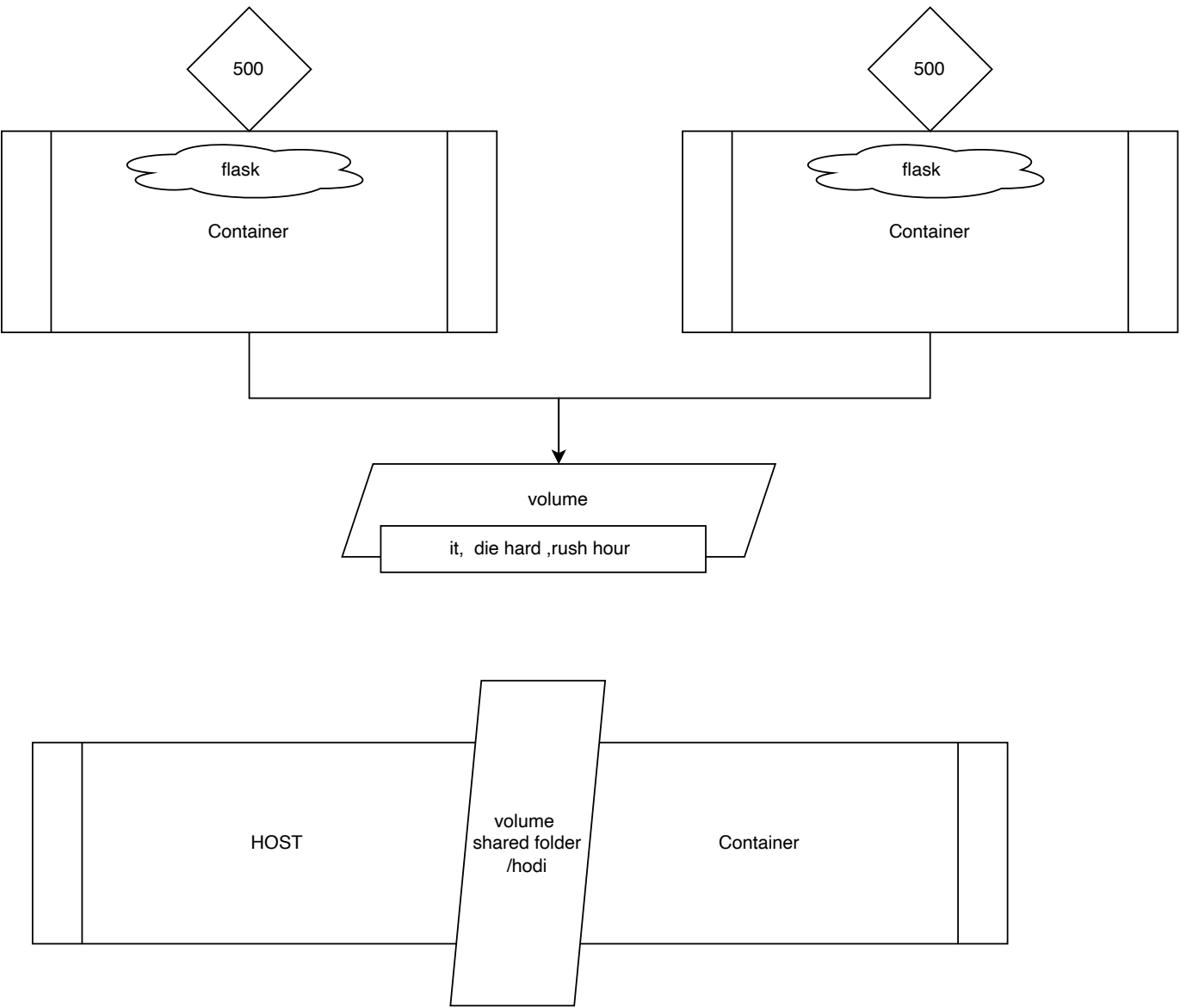
Contianer

Linux

Contianer

MAC

volumes in docker

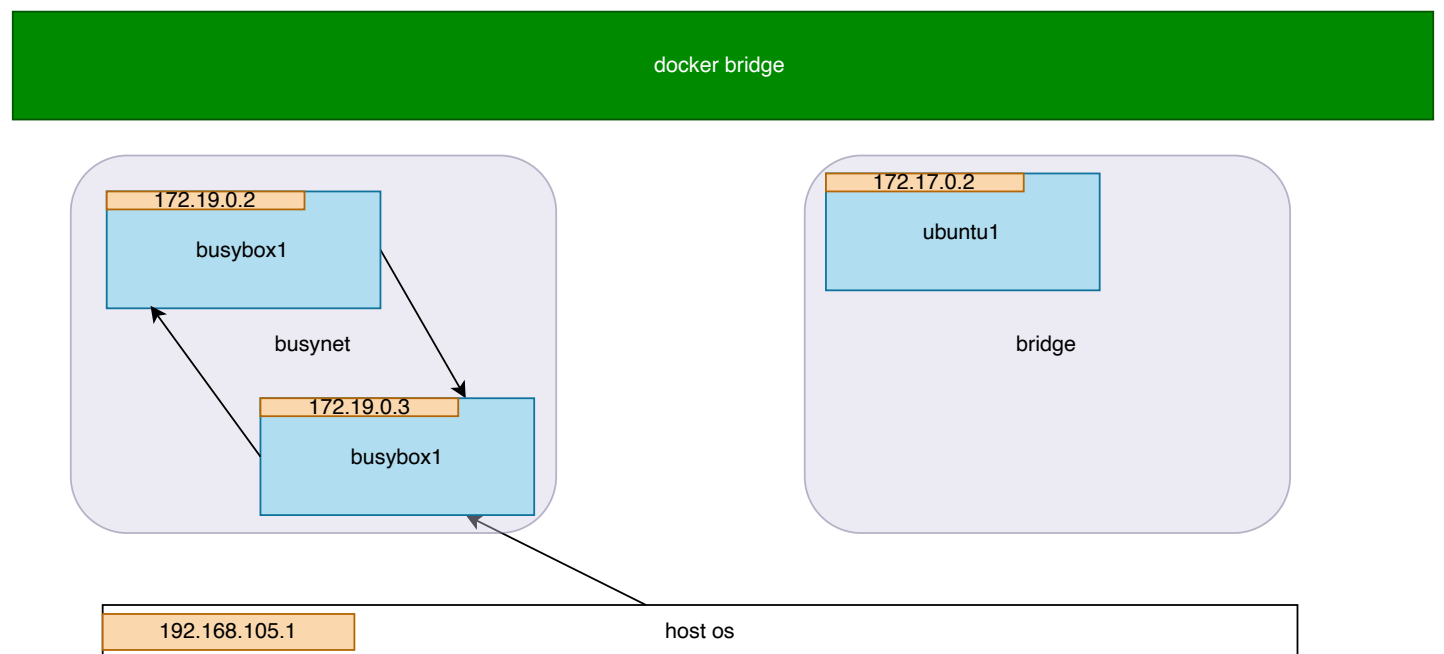
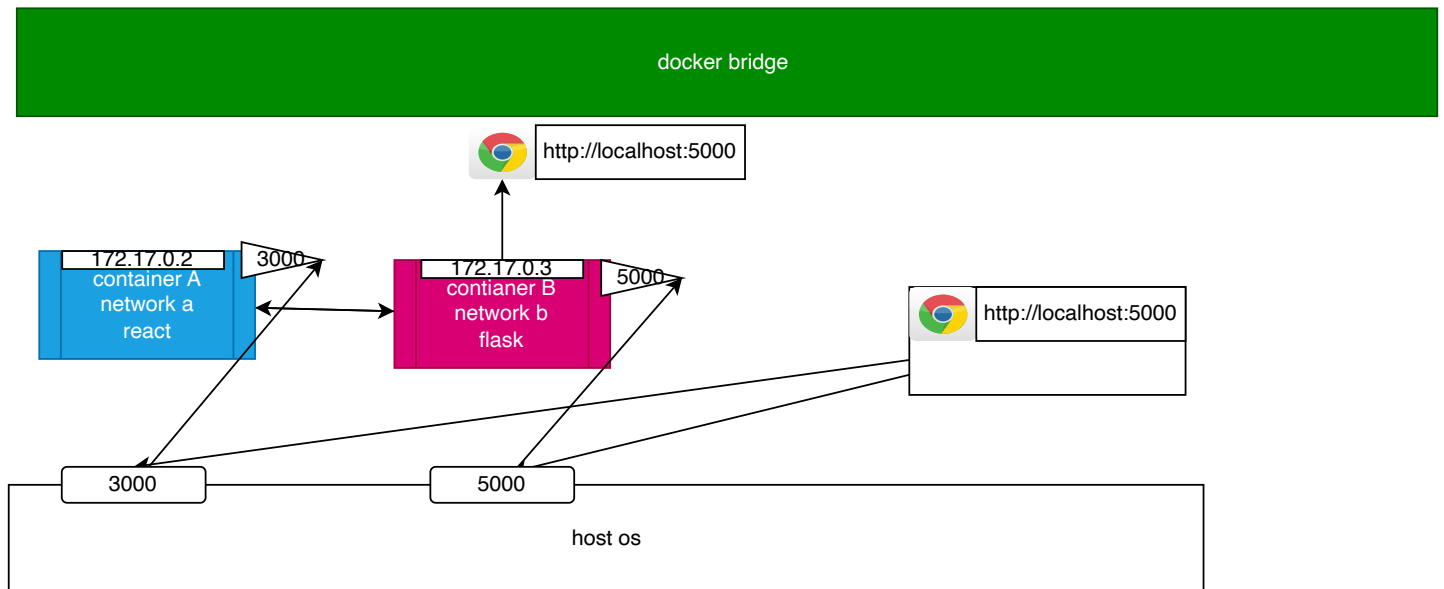


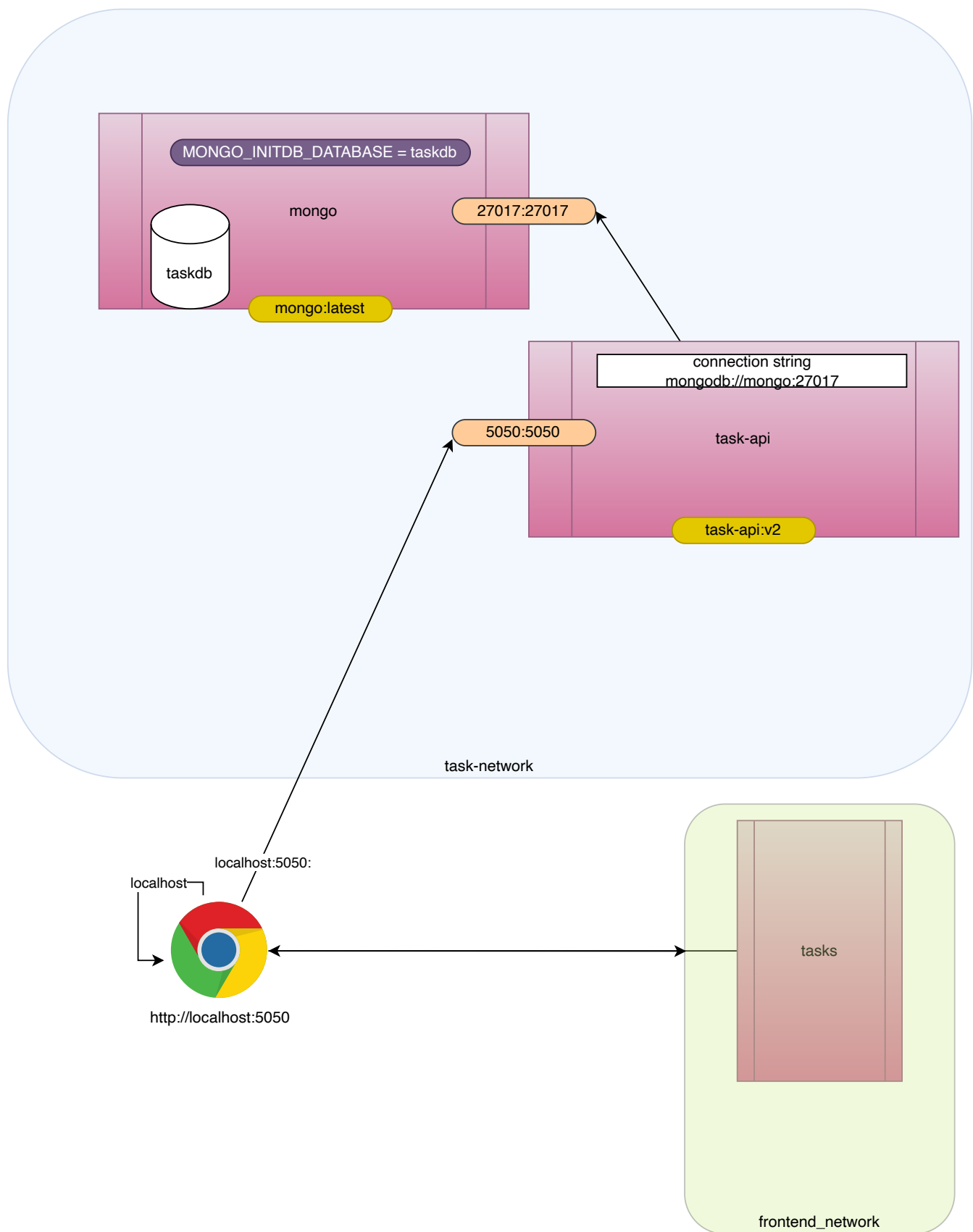
1.1.1.1/24

1.1.1.1/16

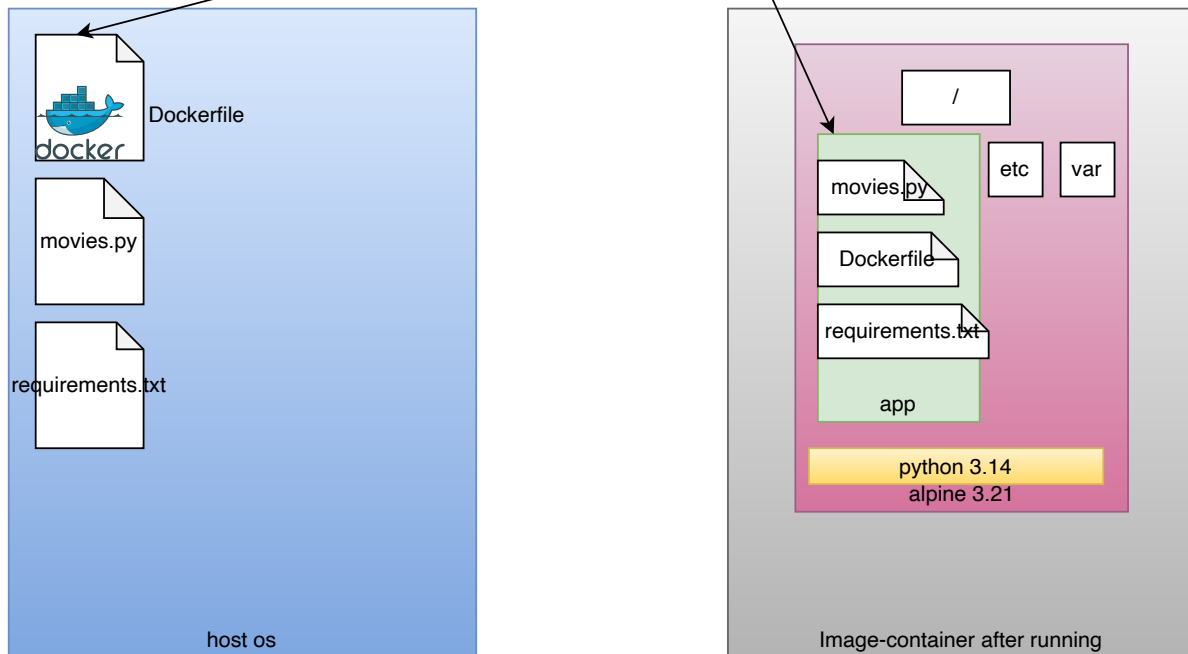
1.1.1.1/32

network name	machine ip
-----------------	---------------

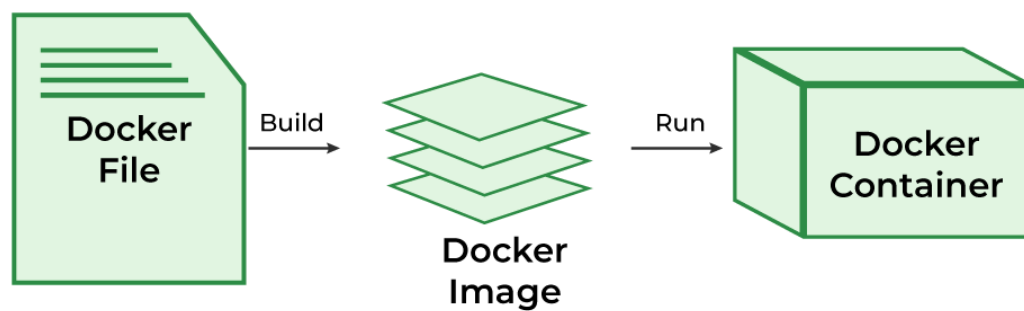
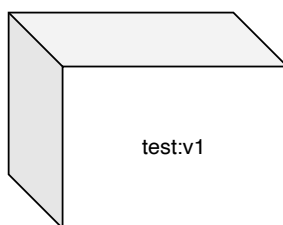




COPY . .



```
docker build -t test:v1
```

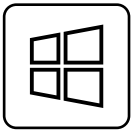
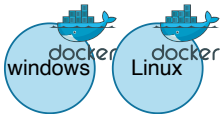


containers

client

REST api

server



virtual machine

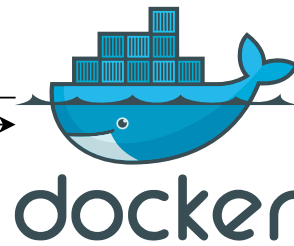


Docker demo

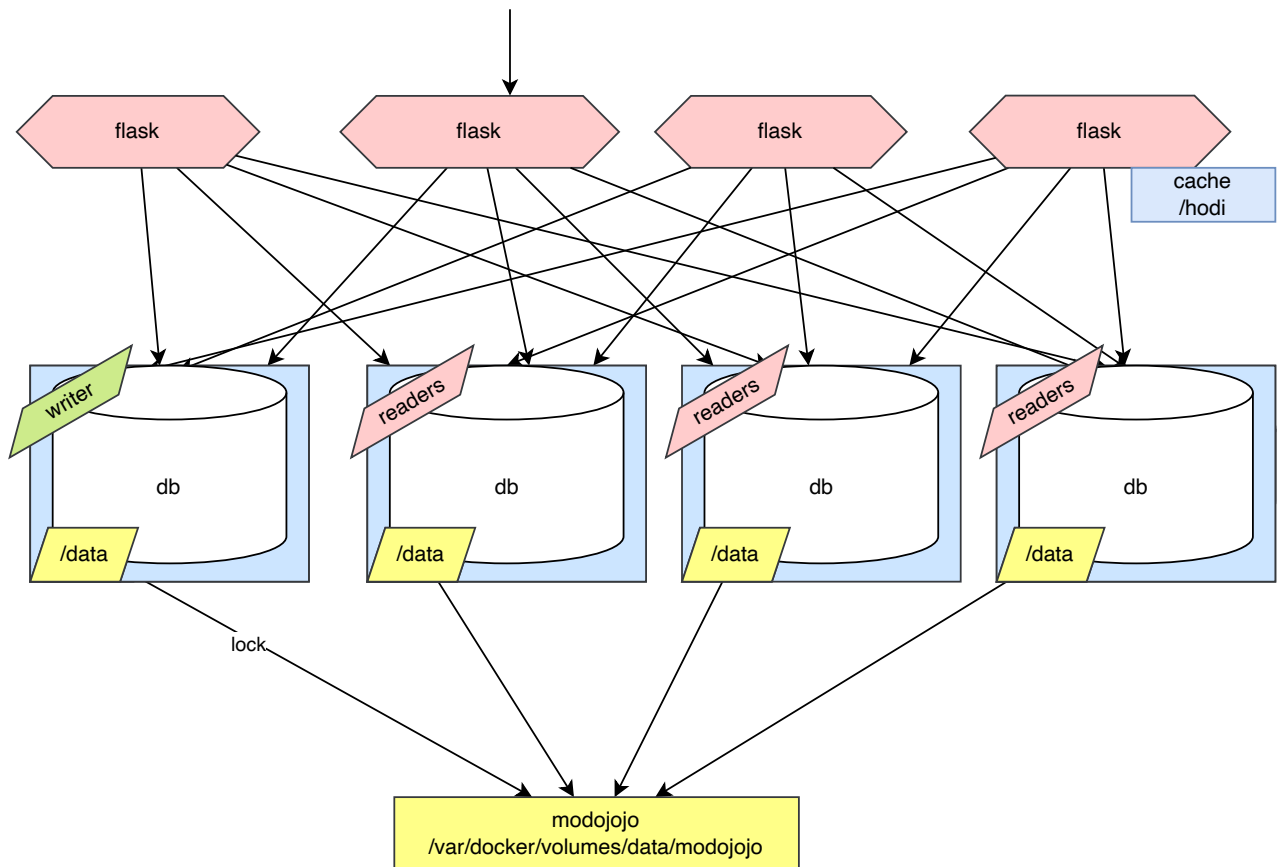
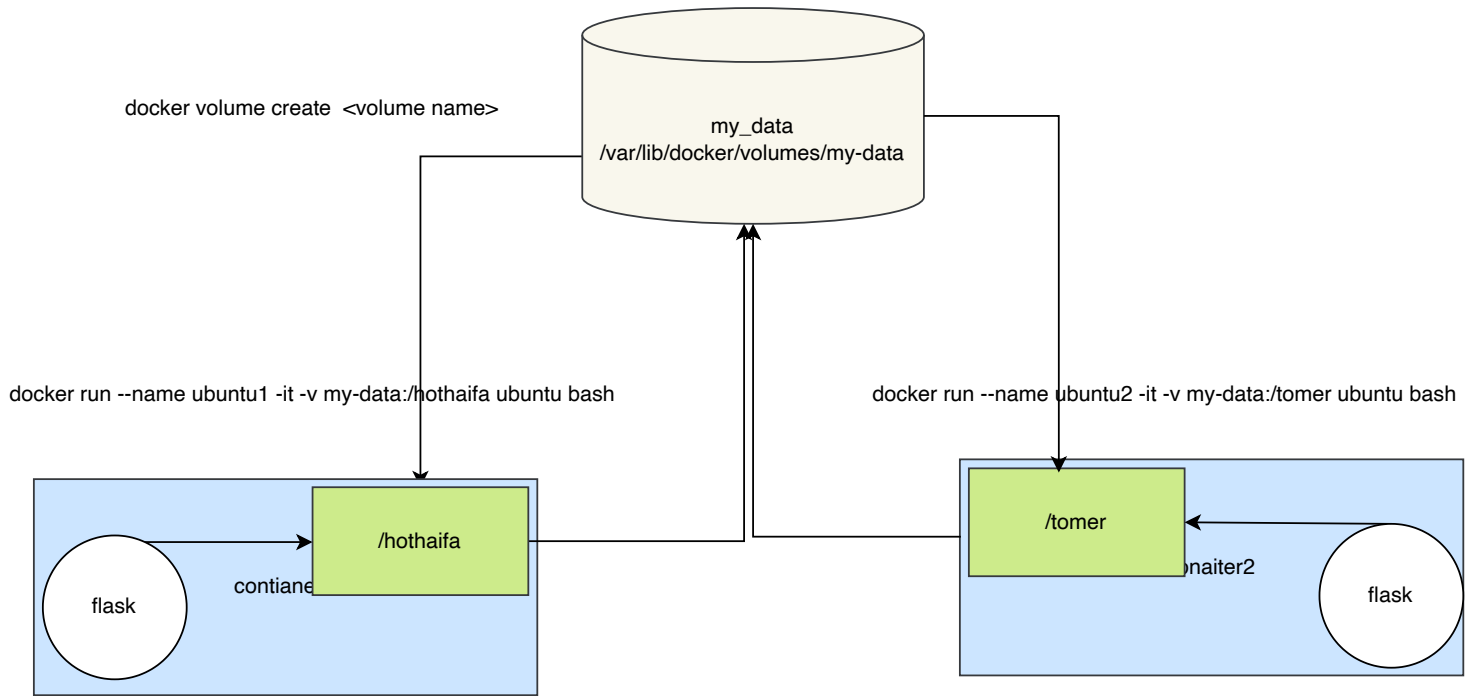
CLI

docker --version

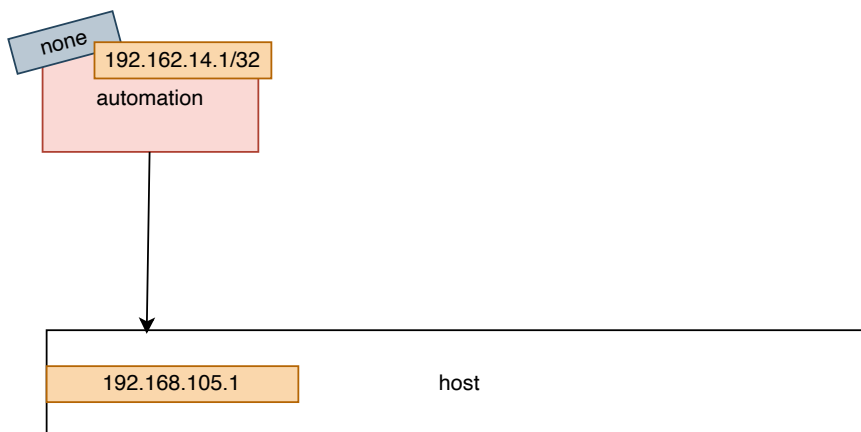
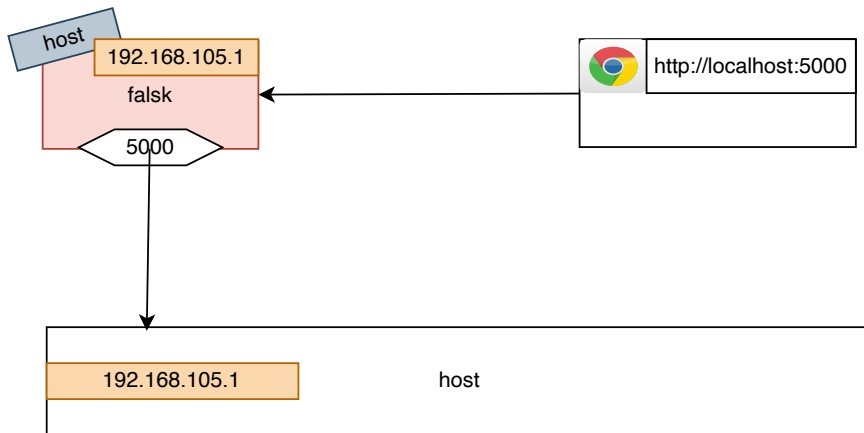
docker pull nginx

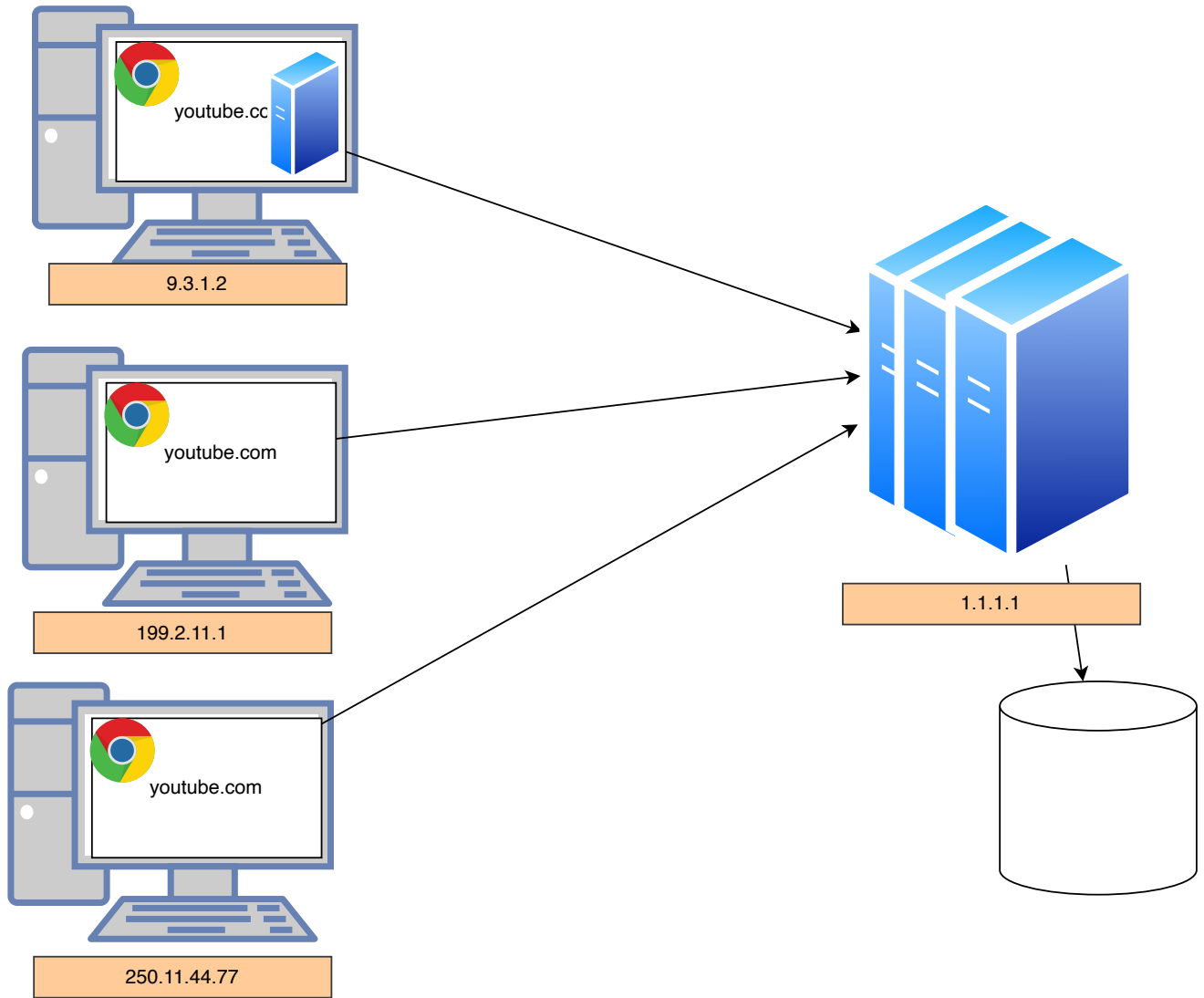


docker hub



docker Host network





queue

